Practical 08: Toolbar and Options Menu

Source: MODULE 2.pdf | Source pages: 1–6

Aim

To develop and test an Android application demonstrating toolbar and options menu.

Steps to Perform the Practical

1. Open Android Studio and create a new project named ToolbarMenuDemo using Empty Views Activity.

2. Open activity_main.xml and add the Toolbar and TextView shown in the practical.

3. Create res/menu/menu_main.xml and add Home, Settings, About and Logout menu items.

4. Open MainActivity.java, set the Toolbar as the support action bar, inflate the menu and handle item selections with Toast messages.

5. Run the app on an emulator or Android device.

6. Tap the three-dot menu and select each option. Confirm the matching Toast message appears.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

Menus, Dialogs and UI Interaction

1. Develop an Android application implementing AppBar (Toolbar) and Options Menu.

Project Name: ToolbarMenuDemo

FileName: activity_main.xml

<?xml version="1.0" encoding="utf-8"?>

```
<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"

   android:id="@+id/main"

   android:layout_width="match_parent"

   android:layout_height="match_parent" android:orientation="vertical">

   <!-- Toolbar -->

   <androidx.appcompat.widget.Toolbar

       android:id="@+id/toolbar"

       android:layout_width="match_parent"

       android:layout_height="?attr/actionBarSize"

       android:background="?attr/colorPrimary"

       app:title="Toolbar Demo"

       app:titleTextColor="@android:color/white"/>

   <!--Main Content -->

   <TextView

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="AppBar and Options Menu Example"
```

```
        android:textSize="22sp"

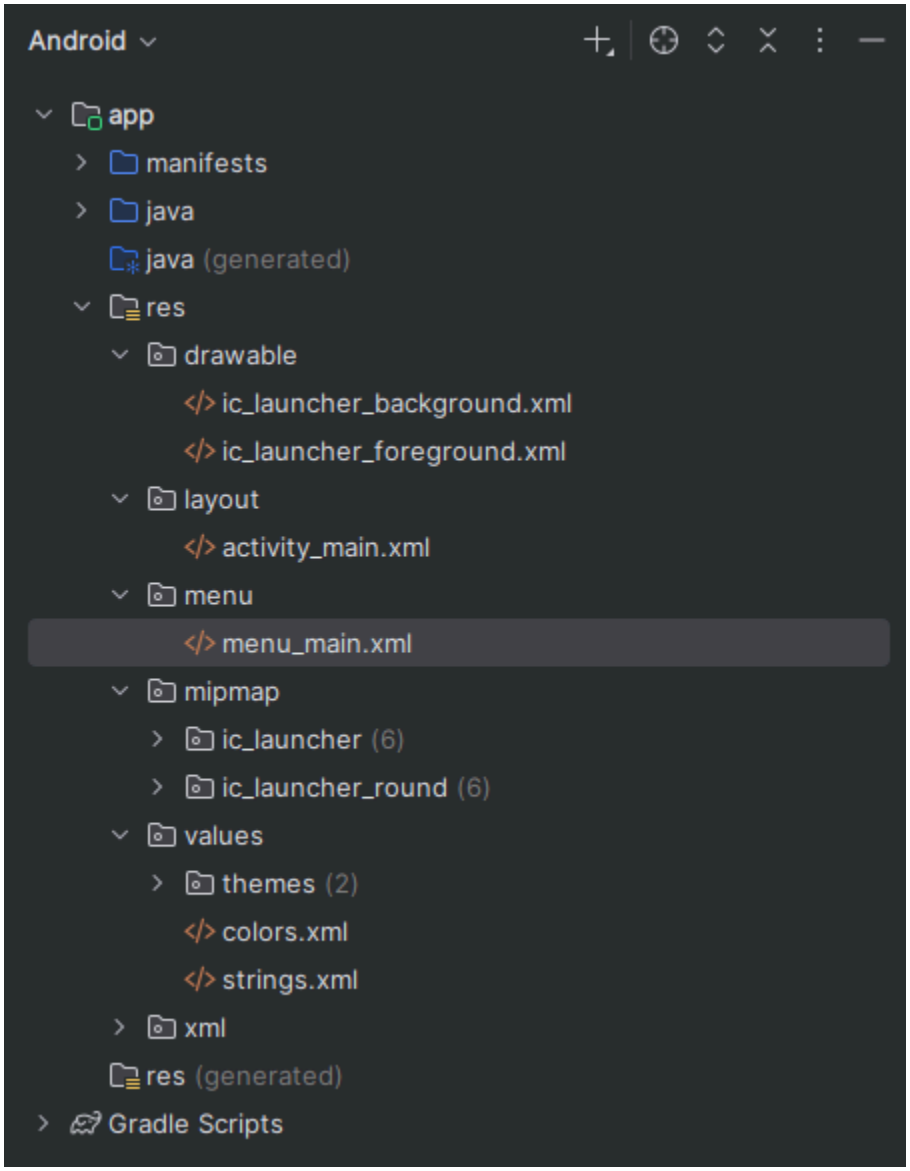
        android:padding="20dp"/>

</LinearLayout>
```

Create a Menu Resource File.
Right Click: Res → New→ Android Resource Directory

Directory Name: menu
Resource Type: menu

Create Menu XML File.
Right Click: menu New→ Menu Resource File Give name as **menu_main.xml**



File Name : menu_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<menu xmlns:android="http://schemas.android.com/apk/res/android"

   xmlns:app="http://schemas.android.com/apk/res-auto">



   <item

       android:id="@+id/menu_home"

       android:title="Home"

       app:showAsAction="never"/>



   <item

       android:id="@+id/menu_settings"

       android:title="Settings"

       app:showAsAction="never"/>

   <item

       android:id="@+id/menu_about"

       android:title="About"

       app:showAsAction="never"/>

   <item

       android:id="@+id/menu_logout"

       android:title="Logout"

       app:showAsAction="never"/>

</menu>
```

File Name: MainActivity.java.

```
package com.example.toolbarmenudemo;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;

public class MainActivity extends AppCompatActivity {

   Toolbar toolbar;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);
       toolbar = findViewById(R.id.toolbar);
       // Set Toolbar as AppBar
       setSupportActionBar(toolbar);
   }
   // Create Options Menu
   @Override
   public boolean onCreateOptionsMenu(Menu menu) {
```

```
        getMenuInflater().inflate(R.menu.menu_main, menu);

        return true;

    }

    // Handle Menu Item Click

    @Override

    public boolean onOptionsItemSelected(@NonNull MenuItem item) {

        int id = item.getItemId();

        if (id == R.id.menu_home) {

            Toast.makeText(this, "Home Selected", Toast.LENGTH_SHORT).show();

        } else if (id == R.id.menu_settings) {

            Toast.makeText(this, "Settings Selected",
Toast.LENGTH_SHORT).show();

        } else if (id == R.id.menu_about) {

            Toast.makeText(this, "About Selected", Toast.LENGTH_SHORT).show();

        } else if (id == R.id.menu_logout) {

            Toast.makeText(this, "Logout Selected", Toast.LENGTH_SHORT).show();

        }

        return true;

    }

}
```


Toolbar Demo
AppBar and Options Menu Example